

CONCORDIA BLOSSER, KS, USA

WMO: 724580

Lat: 39.551N

Lon: 97.651W

Elev: 1469

StdP: 13.93

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13984

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.0	6.4	-6.9	4.0	4.1	-2.5	5.1	8.6	28.4	31.5	26.1	34.4	10.4	350	0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	98.3	74.0	95.1	73.8	92.1	73.2	77.8	90.4	76.5	89.1	75.1	87.6	13.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	136.1	84.2	72.8	128.7	82.8	71.4	122.6	81.4	42.4	90.5	40.9	89.0	39.6	87.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.9	24.4	21.3	-5.0	103.1	3.8	2.8	-7.7	105.1	-10.0	106.7	-12.1	108.3	-14.9	110.3
			-5.7	80.3	3.7	1.9	-8.3	81.7	-10.5	82.8	-12.6	83.9	-15.2	85.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	29.0	32.6	43.4	53.3	63.7	74.6	79.1	76.6	68.9	56.0	42.9	31.5
	DBStd	20.01	11.59	12.46	11.88	9.70	8.42	6.83	6.16	6.10	8.57	9.44	10.08	10.73
	HDD50	2377	653	497	269	74	4	0	0	0	2	47	254	577
	HDD65	5260	1117	906	672	364	129	9	1	1	57	303	662	1039
	CDD50	3994	1	11	65	174	429	738	904	824	569	235	42	2
	CDD65	1399	0	0	2	14	88	297	439	360	174	25	0	0
	CDH74	15077	0	2	39	188	901	3187	5036	3715	1745	256	8	0
	CDH80	6931	0	0	5	44	308	1461	2587	1744	714	67	1	0
	WSAvg	10.6	10.3	10.9	11.9	12.6	10.9	10.8	9.5	8.8	10.1	10.6	10.8	10.1
	PrecAvg	28.8	0.6	0.8	1.9	2.4	4.2	4.3	3.8	3.5	2.8	2.0	1.1	0.9
Precipitation	PrecMax	44.5	1.9	2.5	8.3	6.0	9.7	14.1	16.7	10.7	8.5	5.3	4.9	3.6
	PrecMin	14.5	0.0	0.0	0.0	0.4	0.3	1.0	0.1	0.4	0.2	0.0	0.0	0.0
	PrecStd	7.0	0.4	0.7	1.6	1.2	2.3	2.3	3.0	2.0	1.9	1.4	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.7	70.3	78.6	85.4	93.0	98.9	103.5	101.7	96.6	87.5	74.1	62.7
		MCWB	47.3	51.7	58.4	63.1	69.1	73.9	73.9	72.9	70.9	66.9	57.4	52.0
	2%	DB	54.8	62.7	72.9	79.2	87.5	95.1	99.2	97.1	91.7	80.5	68.2	56.2
		MCWB	43.4	49.2	56.5	61.4	68.1	73.3	74.5	74.3	71.9	63.2	54.8	47.1
	5%	DB	50.0	56.5	67.7	74.4	83.0	92.0	95.7	93.1	87.9	75.9	63.4	51.3
		MCWB	41.1	46.3	54.3	59.6	67.2	72.6	74.6	74.3	70.2	61.5	52.6	43.2
	10%	DB	45.2	50.9	62.5	70.2	79.0	88.5	92.4	89.3	83.8	71.8	58.7	46.9
		MCWB	38.5	42.4	51.8	57.5	65.9	71.7	74.3	73.4	68.3	59.7	50.0	40.2
	0.4%	WB	49.0	56.0	63.8	67.4	73.8	78.2	80.3	79.8	75.8	71.1	60.5	55.4
		MCDB	57.8	63.4	73.2	78.4	85.7	90.4	92.6	91.3	88.0	80.9	67.6	58.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.5	51.0	60.2	64.4	71.5	76.2	78.2	77.5	73.8	67.5	57.5	48.2
		MCDB	53.1	60.7	67.3	74.0	82.4	89.7	91.1	89.5	86.0	75.3	64.0	53.9
	5%	WB	41.6	46.5	56.5	61.8	69.5	74.6	76.9	76.0	72.1	64.3	54.0	43.8
		MCDB	49.4	56.1	65.9	72.4	79.5	87.6	89.8	87.9	84.0	72.2	61.9	50.7
	10%	WB	38.6	42.7	52.1	58.8	67.4	73.1	75.4	74.5	70.4	61.2	50.5	40.6
		MCDB	45.2	50.6	61.5	68.9	76.6	85.3	88.4	85.9	81.0	69.7	58.6	46.8
Mean Daily Temperature Range	5% DB	MDBR	18.9	20.0	22.5	22.3	21.3	21.9	21.5	21.3	22.4	22.4	20.5	18.1
		MCDBR	26.6	28.8	29.5	28.5	26.5	25.3	25.3	25.4	25.2	27.6	26.3	25.3
		MCWBR	17.2	17.8	16.5	15.8	12.1	9.5	8.6	9.0	9.7	14.1	15.5	16.9
	5% WB	MCDBR	25.2	26.9	25.4	25.2	23.1	23.0	22.2	21.7	22.3	22.2	23.5	23.9
		MCWBR	16.9	17.7	15.8	15.9	12.1	10.2	9.6	9.9	9.9	13.2	15.6	17.0
Clear-Sky Solar Irradiance	taub		0.282	0.292	0.324	0.358	0.386	0.399	0.415	0.412	0.373	0.331	0.287	0.275
	taud		2.491	2.466	2.411	2.345	2.344	2.348	2.323	2.330	2.385	2.483	2.530	2.506
	Ebn at Noon		285	297	296	291	283	279	273	271	276	278	281	279
	Edh at Noon		24	29	34	38	39	39	40	39	34	28	23	22
All-Sky Solar Radiation	RadAvg		739	990	1320	1659	1908	2161	2129	1881	1547	1122	819	633
	RadStd		59	85	98	92	129	103	128	101	89	111	63	62

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page